

Part1

Hello, my name is _____,

Thank you _____ for participating in our study. We aim to gather insights into testing for and measuring discriminatory behavior in the outcome of software systems. We look forward to utilizing your insights to further our research.

Now, we will review the consent form included in the interview sign-up survey.

[\[Revisit the consent form and open for questions\]](#)

This interview will be recorded for internal purposes only. Additionally, we will be taking notes during the interview. The consent form that was sent to you specifies that your identity will remain confidential. Other than your email address, no other personal information will be collected. Your email will be used only to schedule your interview and compensation. You can withdraw the study at any point. Are you still willing to participate in this study?

If you get disconnected from the call, please use the same link included in the email invitation, join via phone, or email if you need to reschedule.

Before we begin, do you have any questions for me?

I'll begin the interview recording now.

[Start Recording]

Background Questions

1. To begin, tell us a little about yourself.
2. What is your current role/occupation?
 - a. How familiar are you with traditional software testing methodologies? Maybe you can share the kind of experience in software testing with us.

Interview Questions

Thank you for responding! Now I want to talk more about your experiences & expectations in understanding discriminatory behavior in software systems.

******* Part 1*******

Section: Experiences with Discrimination in Software Outcomes

1. Can you describe an experience in the software engineering field?
2. Do you have any experience in encountering discriminatory software behavior?
3. When you recognized that potential for discriminatory behavior, what was your *expectation* of how the system should have behaved instead?

Part2

*****Part 2*****

Thank you for sharing your experiences and perceptions regarding software discrimination. Next, we would like to get your perception on the utility and usability of different approaches to create unit tests for measuring and understanding discriminatory behavior in software outcomes. For this study we are considering two different test generation tools. First is Pynguin which is an automated unit test generation for Python. Second is Themis, an automated causal unit test generation tool.

Now I will elaborate for you 2 separate scenarios. I want you to open our project and do the following tasks according to instructions, which you will have access to as you are working

Section 1: Themis Run

Scenario 1: This is a simple loan application system which is a Rule-based software loan.py. The behavior of this software is quite deterministic. We aim to understand the user perspective on understanding discriminatory behavior of such systems through unit test generation approaches. Similarly an ML based program which is probabilistic.

- **Input Specification Task**

- A. Scenario: Given the loan approval case, participants define input attributes (e.g., gender, race, income) as categorical.

- C. Through Themis generate unit tests running grid3_2_0.py

- D. look at the results and go to the following question

- E. Similarly run [grid3.py](#) and generate result for ML based program and answer the following question

- **Think-Aloud Prompt**

1. Did you face any challenges during setting up the run environment
2. What do you think these unit tests are verifying/communicating for both of the programs? Any challenges understanding them?
3. Did you face any challenge during test generation or understanding generated tests?

4. Do you think the generated unit tests help you reason about how the program shows discrimination?

- If so, why?
- If not, why not?

4. Based on these results, do you notice any potential next action you could take upon?

Section 2— Comparison Deterministic vs Non deterministic(only for Themis):

Scenario 2: In this next scenario, we will test a Telecom Churn prediction software that uses ML models. We want users to understand discrimination in such a non deterministic ML-based software outcome. The pseudocode is in slide 11.

- **Input Specification Task**

A. Scenario: Given churn prediction dataset in slide 12 and as the input subspace is huge we aimed to use a subset of input space which is shown in slide 12, participants define input attributes (e.g., gender, SeniorCitizen, partner, dependents, tenure) as categorical and other test configurations are similar to loan application software.

B. Generate automated test suite through Themis which is in slide 13 and the result is in slide 14.

C. Penguin generated tests are shown in slide 15.

- **Think-Aloud Prompt**

What do you think these test cases are designed to verify about the ML model(in slide 13 and 15)? *(Open-ended) in*

From these tests, can you infer any patterns of bias or discrimination?
(Open-ended)

Q1. What do you find useful about Themis's approach of generating repetitive testcase for non-deterministic systems(ML Churn Prediction), where it may produce different results for the same test case?*(open ended)*

Q2. How can or should discrimination-oriented testing methodology differ between rule-based and ML-based decision making systems?*(Open-ended)*

Q3. Did these generated tests help you to decide to make any changes to the program/dataset to mitigate discrimination in outcome? If yes, How so?*(Open-ended)*

Q4. Do you find altering more than one attribute can be more insightful in slide 16 and 17? Do you think altering more than one value can provide more deeper insight in understanding causal discrimination.

Q5. Do you have any recommendations for improving causal testing methodology for understanding discrimination in software outcomes?

Part3

*****Part3*****

LLM has been identified as a promising tool for unit test generation. We want to identify if LLM can play a role in such discrimination testing. For this we considered ChatGPT.

Prompt: We provided you with a prompting template for the user to write prompts for analyzing such kinds of discrimination. Feel free to change anything in the prompting structure

Section 2: GPT Run

Scenario 1: This is a simple loan application system which is a Rule-based software for llm which is `themis_llm_rulebased.py`. The behavior of this software is quite deterministic. We aim to understand the user perspective on understanding discriminatory behavior of such systems through unit test generation approaches.

- **Input Specification Task**

A. Scenario: Given the loan approval case, participants define input attributes (e.g., gender, race, income) as categorical.

C. Through Themis generate unit tests running `themis_llm_rulebased.py`

D. Look at the results and go to the following question

E. Similarly `ml_llm_audit.py` and generate result for ml program and answer the following question

- **Think-Aloud Prompt**

1. Did you face any challenges during setting up the run environment or prompt refinement?
2. What do you think these unit tests are verifying/communicating for both of the programs? Any challenges understanding them?
3. Did you face any challenge during test generation or understanding generated tests?

4. Do you think the generated unit tests help you reason about how the program shows discrimination?
 - If so, why?
 - If not, why not?

4. Based on these results, do you notice any potential next action you could take upon?